

Chapter 05: Power Series Solutions about Ordinary Points

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Course: Applied Ordinary Differential Equations (ENGR 213) | **Textbook:** Dennis G. Zill (7th Ed., Section 5.1.2)

1. Executive Summary & First-Principles Philosophy

When differential equations have variable coefficients (such as Airy's equation $y'' - xy = 0$ or Bessel's equation), closed-form elementary solutions (polynomials, sines, exponentials) generally do not exist.

Instead, we represent the solution as an infinite **Power Series** centered at an ordinary point x_0 :

$$y(x) = \sum_{n=0}^{\infty} c_n (x - x_0)^n$$

Substituting this series and its derivatives into the ODE allows us to shift summation indices, combine like powers of x , and derive a **Recurrence Relation** that expresses all higher coefficients $c_2, c_3, c_4, \dots$ in terms of the two fundamental arbitrary constants c_0 and c_1 .

2. Core Mechanics & Mathematical Engine

A. Ordinary vs Singular Points

Given standard form $y'' + P(x)y' + Q(x)y = 0$:

- A point x_0 is an **Ordinary Point** if both coefficient functions $P(x)$ and $Q(x)$ are **analytic** at x_0 (meaning they possess a convergent Taylor series; for rational functions, denominators must be non-zero at x_0).
- If either $P(x)$ or $Q(x)$ fails to be analytic at x_0 , the point is a **Singular Point**.

B. Index Shifting Mechanics

The key computational skill is shifting the summation index so that all sums share the common power x^k :

$$\begin{aligned} \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} &\xrightarrow{k=n-2} \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k \\ \sum_{n=1}^{\infty} n c_n x^n &\xrightarrow{k=n} \sum_{k=1}^{\infty} k c_k x^k \end{aligned}$$

3. Fully Worked Exam Archetype: Airy's Equation ($y'' - xy = 0$)

Problem: Find the first 4 non-zero terms of the general power series solution about $x_0 = 0$ for:

$$y'' - xy = 0$$

Step-by-Step Solution:

- **Step 1: Assume power series representations:**

$$y = \sum_{n=0}^{\infty} c_n x^n, \quad y' = \sum_{n=1}^{\infty} n c_n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

- **Step 2: Substitute into ODE:**

$$\begin{aligned} \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x \sum_{n=0}^{\infty} c_n x^n &= 0 \\ \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} &= 0 \end{aligned}$$

- **Step 3: Shift indices to common power x^k :**

- First sum: let $k = n - 2 \implies n = k + 2$. Starts at $k = 0$:

$$\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$$

- Second sum: let $k = n + 1 \implies n = k - 1$. Starts at $k = 1$:

$$\sum_{k=1}^{\infty} c_{k-1} x^k$$

- **Step 4: Peel off the $k = 0$ term from the first sum:**

$$(2)(1) c_2 x^0 + \sum_{k=1}^{\infty} [(k+2)(k+1) c_{k+2} - c_{k-1}] x^k = 0$$

- **Step 5: Set coefficients to zero:**

$$2c_2 = 0 \implies c_2 = 0$$

$$(k+2)(k+1) c_{k+2} - c_{k-1} = 0 \implies c_{k+2} = \frac{c_{k-1}}{(k+2)(k+1)}, \quad k \geq 1$$

- **Step 6: Compute coefficients iteratively:**

- For $k = 1$: $c_3 = \frac{c_0}{(3)(2)} = \frac{c_0}{6}$
- For $k = 2$: $c_4 = \frac{c_1}{(4)(3)} = \frac{c_1}{12}$
- For $k = 3$: $c_5 = \frac{c_2}{(5)(4)} = 0$ (since $c_2 = 0$)

- For $k = 4$: $c_6 = \frac{c_3}{(6)(5)} = \frac{c_0/6}{30} = \frac{c_0}{180}$

- **Step 7: Assemble two linearly independent solutions:**

$$y(x) = c_0 \left(1 + \frac{x^3}{6} + \frac{x^6}{180} + \dots \right) + c_1 \left(x + \frac{x^4}{12} + \dots \right)$$

4. Rapid Exam Traps & Red Flags

- **⚠️ Trap 1: Peeling Off Terms:** Never combine sums before aligning their starting summation indices! If one sum starts at $k = 0$ and the other at $k = 1$, evaluate $k = 0$ separately before combining.
- **⚠️ Trap 2: Losing the Distinction between c_0 and c_1 :** The general solution to a 2nd-order ODE must have exactly two independent arbitrary constants (c_0 and c_1). Every other coefficient c_n must be expressed purely in terms of c_0 or c_1 .